

Design Journal 1

DESN1000 - Week 4

UNSW zID: *xxxxxxx*

Page Count: 48

Contents

1	Design Process	1
1.1	Week 1	1
1.1.1	Entry 1 (18/02)	1
1.1.2	Entry 2 (20/02)	2
1.1.3	Entry 3 (21/02)	4
1.2	Week 2	4
1.2.1	Entry 4 (23/02)	4
1.2.2	Entry 5 (26/02)	6
1.2.3	Entry 6 (27/02)	8
1.3	Week 3	10
1.3.1	Entry 7 (02/03)	10
1.3.2	Entry 8 (05/03)	14
1.3.3	Entry 9 (07/03)	16
1.4	Week 4	17
1.4.1	Entry 10 (09/03)	17
1.4.2	Entry 11 (12/03)	19
2	Teamwork	20
2.1	Week 1	20
2.1.1	Entry 1 (18/02)	20
2.1.2	Entry 2 (20/02)	21
2.1.3	Entry 3 (21/02)	23
2.2	Week 2	24
2.2.1	Entry 4 (23/02)	24
2.2.2	Entry 5 (26/02)	25
2.2.3	Entry 6 (27/02)	26
2.3	Week 3	28
2.3.1	Entry 7 (02/03)	28
2.3.2	Entry 8 (05/03)	30
2.3.3	Entry 9 (07/03)	32
2.4	Week 4	32
2.4.1	Entry 10 (09/03)	32
2.4.2	Entry 11 (12/03)	34
3	Reflections	35
3.1	Week 1	35
3.1.1	Entry 1 (18/02)	35
3.1.2	Entry 2 (20/02)	36
3.1.3	Entry 3 (21/02)	37
3.2	Week 2	38
3.2.1	Entry 4 (23/02)	38
3.2.2	Entry 5 (26/02)	39
3.2.3	Entry 6 (27/02)	40

3.3	Week 3	42
3.3.1	Entry 7 (02/03)	42
3.3.2	Entry 8 05/03	43
3.3.3	Entry 9 (07/03)	44
3.4	Week 4	45
3.4.1	Entry 10 (09/03)	45
3.4.2	Entry 11 (12/03)	47

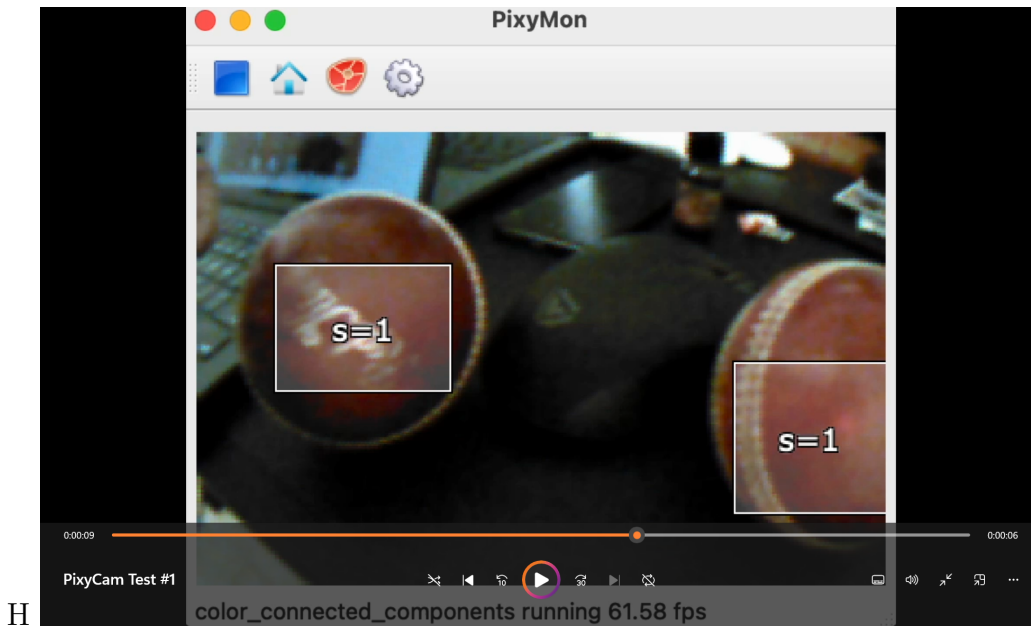


Figure 3: Screenshot of PixyMon running colour_connected_components. Done by me

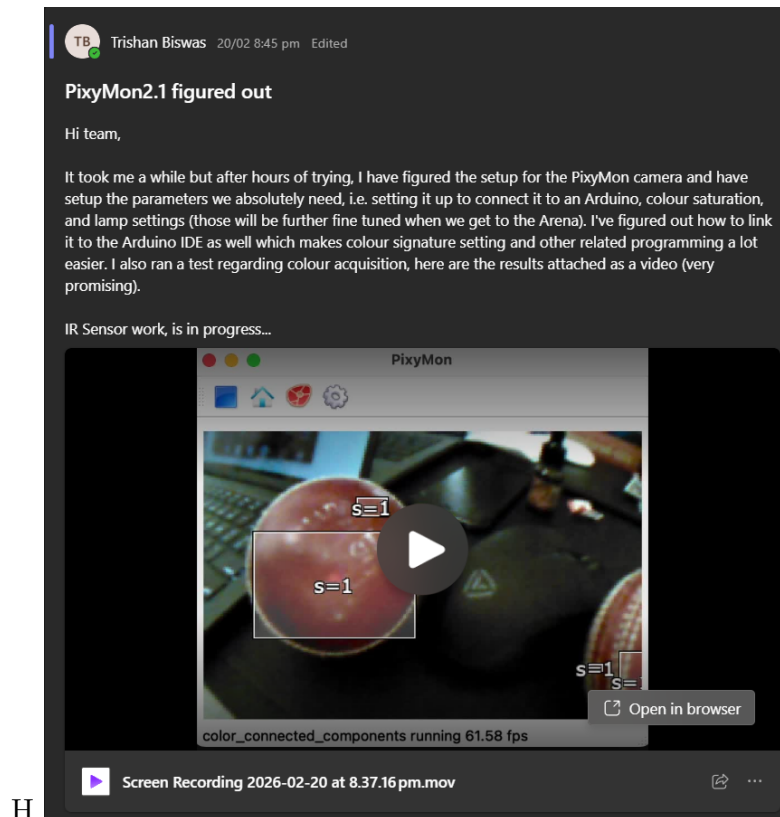


Figure 4: Communication to Team C01 on Microsoft Teams for the update. Done by me

1.1.3 Entry 3 (21/02)

On the 21st of February, I began the action item that I took initiative in setting up regarding setting up a “component to subsystem layout” for all the kit items we are provided with to inform everyone what each component’s role is within the robot system. I approached this by individually mapping key kit components to subsystems and defined its role within the robot.

Subsequently, I utilised LucidSpark to produce a component role mindmap to visualise these component groupings by subsystem to have a clear system overview. This is particularly useful for those people unfamiliar with these components in Team C01.

Ultimately, I think this exercise prefaced the full scope of the robot’s overall design and architecture whilst confirming that no critical parts were missing or unaccounted for. See **Figure 5** below for the LucidSpark mindmap **I created**.

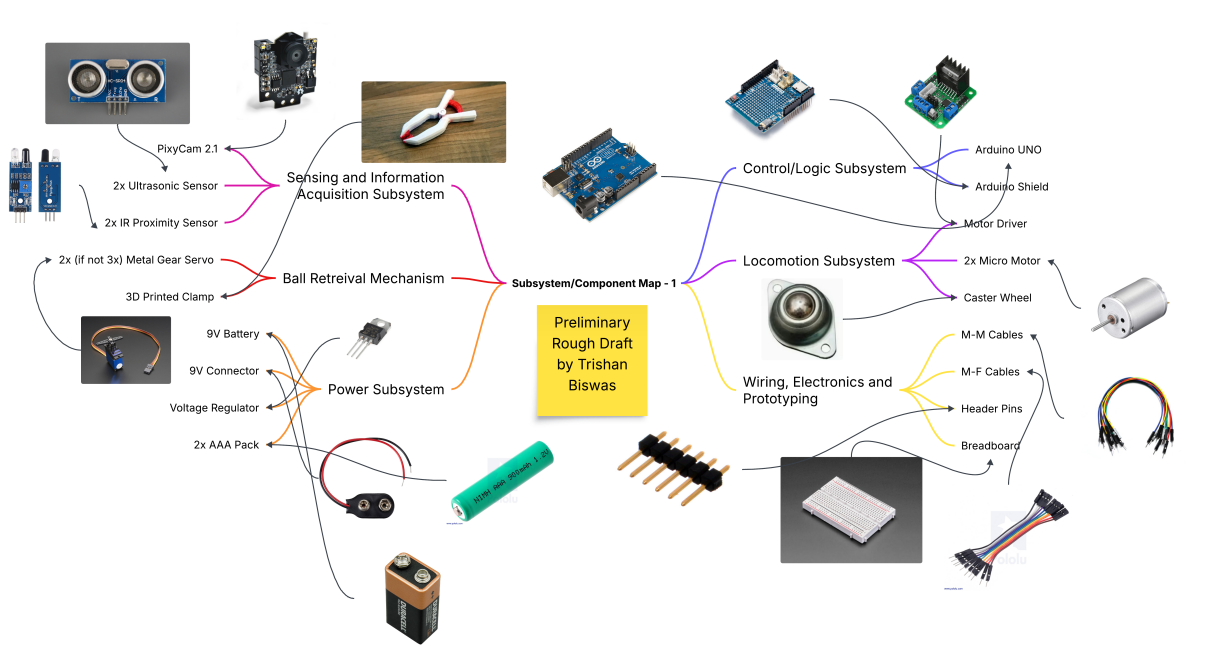


Figure 5: LucidSpark mindmap with subsystem to component layout. **Done by me**

1.2 Week 2

1.2.1 Entry 4 (23/02)

For the very first entry of week 2, I will discuss how I designed a formal sensor testing plan for the IR sensor and the Ultrasonic sensor along with running and designing a simulation of the test on TinkerCAD circuits. This was all done before Week 3 to have a solid plan before running the tests physically in Week 3.

Having run a preliminary PixyCam test, it only makes sense to configure and run tests with the other two sensors too. However, as of now we haven’t been provided with enough

details regarding the functioning of the sensors and how to code with it on the Arduino IDE. Hence, **I did** my own independent research by watching various video tutorials and reading up on important information regarding the setup (i.e. which pin on the ultrasonic sensor means what and where is it connected to, how to program with it and receive distance calculations). The sources I used to learn these are listed in the “Sources” section.

After learning and gathering information regarding the setup, I ran a simulation of the ultrasonic sensor first on TinkerCAD. See **Figure 6** below for the ultrasonic sensor simulation.

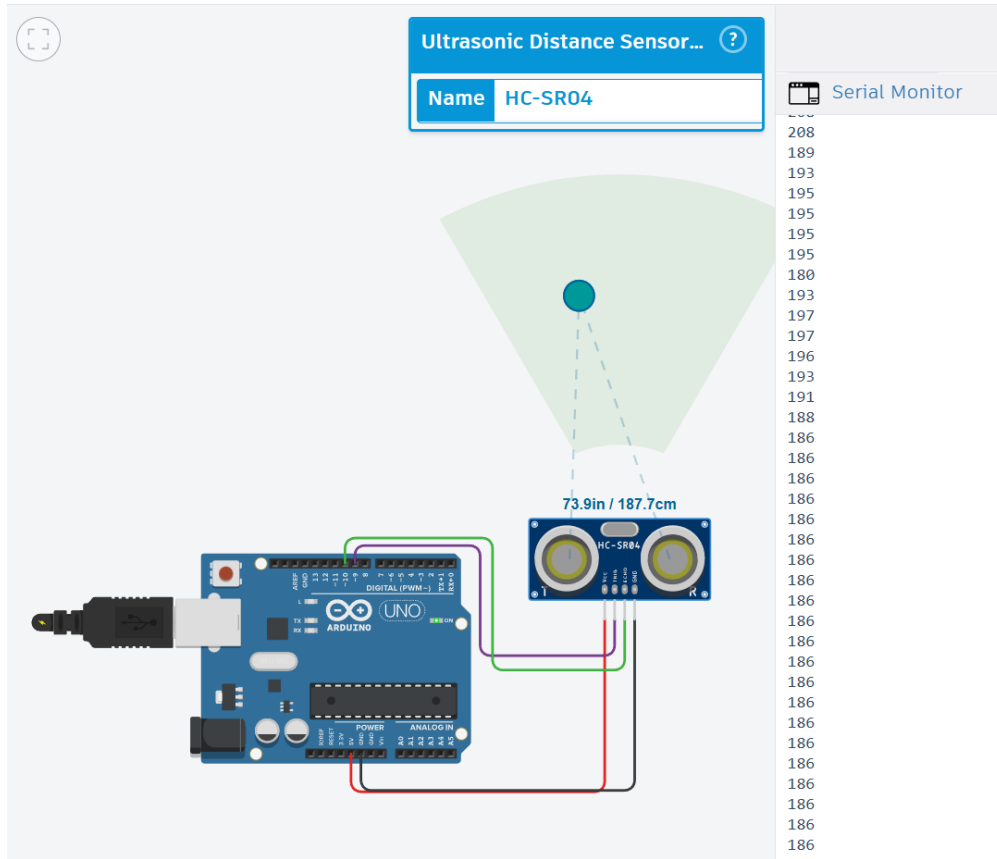


Figure 6: HC-SR04 Sensor simulation ahead of Week 3 physical sensor test. **Made by me**

I then calculated the percent error in measured distance against the actual distance by running multiple tasks, and they all seemed to be off by $\tilde{1.0}$ -1.7cm from the actual measurement. Hence, with the value in **Figure 6**, the percent error was calculated as follows:

$$\frac{187.7 - 186.0}{187.7} \cdot 100\% = 0.91\% \approx 1\%$$

Which is still very accurate and negligible in the use case of ball range finding.

Subsequently, I designed and ran another simulation to test the IR Obstacle/Proximity sensor and verified it's performance by placing an obstacle in front of it and connecting

an LED (with a 200Ω resistor connected to another digital pin which would have a power outputted (to light the LED) by the Arduino when a LOW signal is passed through the IR Sensor (i.e. there is an object detected by the sensor). See **Figure 7** below for the IR sensor simulation.

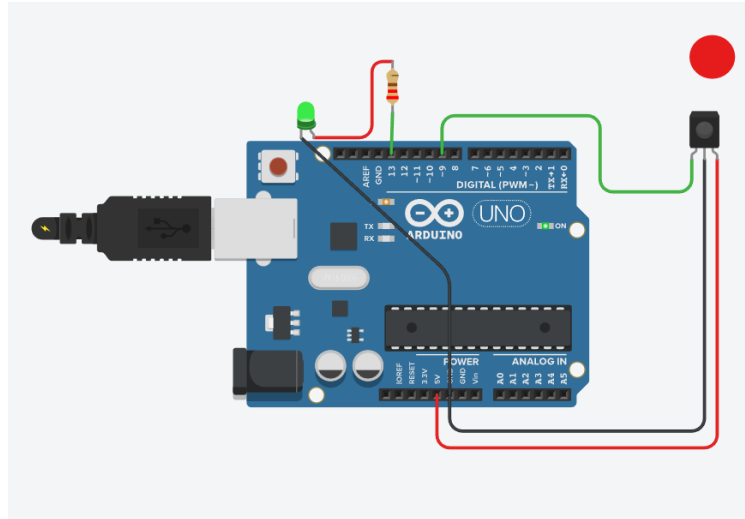


Figure 7: IR Sensor simulation ahead of Week 3 test, green LED is on (object detected). **Made by me**

1.2.2 Entry 5 (26/02)

For entry 2 of week 2, the main agenda from the mechanical subgroup's side was to run a test on the metal gear servo while linked to the Arduino UNO. Hence, on the 26th of Feb the mechanical subgroup leader Franklin and I (programming leader) set up a Microsoft Teams meeting setting up the Arduino UNO, wiring the servo pins to the UNO, and coding the "sweep" on the IDE.

The metal gear servo was taken by Franklin and I assisted in the coding side of things while we were on call. See **Figure 8** for evidence of the meeting.

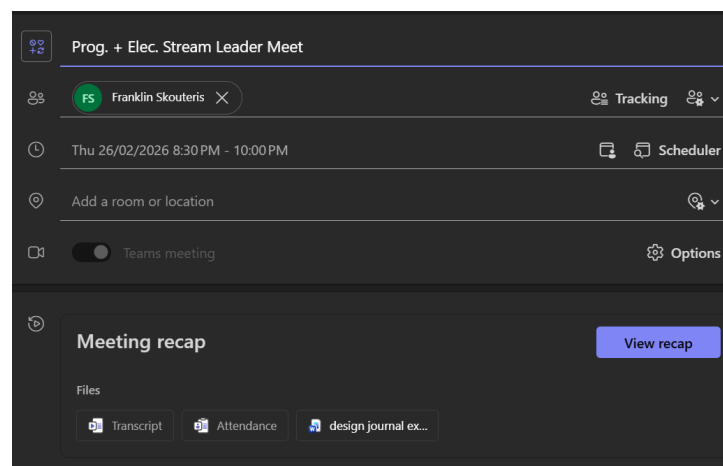


Figure 8: Meeting organised for preliminary servo sweep test. **Organised by me & Franklin**

In this meeting, we wrote and ran a servo “sweep” test directly on the Arduino IDE using one metal gear servo attached appropriately. Then, the code swept the servo through a bunch of angle ranges to determine whether it has a limit. Subsequently, there was, as we initially tested angles beyond 180° and a full 360° rotation, we were expecting a continuous motion or an extended rotation beyond 180° . However, the discovered that the servo has a maximum range of motion (ROM) of 180° . This was a critical finding for the servo aided clamp mechanism design. That is, the claw must be programmed within 0° to 180° range to open and close.

The next steps now are to figure out and map servo sweep angles to an “open” and “close” configuration on the clamp. See **Figure 10** below for an image of the servo sweep code. In it, the servo rotates from 0° to 180° and back in 20° increments, with a 100 ms delay per step. The highlighted rows indicate key configuration lines while the annotations describes the function of each statement.

```
1 // @ Trishan Biswas
2
3 #include <Servo.h>
4
5 Servo myservo; | // Servo object to control a servo motor
6
7 int pos = 0; | // stores current servo position in degrees
8
9 void setup() {
10 myservo.attach(9); | // attaches servo signal wire to pin 9
11 }
12
13 void loop() {
14
15 // Sweep forward:  $0^\circ \rightarrow 180^\circ$ 
16 for (pos = 0; pos ≤ 180; pos += 20) {
17 myservo.write(pos); | // move to position
18 delay(100); | // wait 100 ms per step
19 }
20
21 // Sweep back:  $180^\circ \rightarrow 0^\circ$ 
22 for (pos = 180; pos ≥ 0; pos -= 20) {
23 myservo.write(pos); | // move to position
24 delay(100); | // wait 100 ms per step
```

Figure 9: Servo sweep test sketch, coded by me and Franklin.



Figure 10: Servo sweep test. Image of the physical test. **Taken by Franklin**

1.2.3 Entry 6 (27/02)

For Entry 6, I will walk through how I generated and completed my parts for the Crazy Eights and what each of my eight ideas were for my self assigned subsystem. That is, for the sensor integration subsystem. I generated eight distinct concepts under time limits with rough sketches for each of them. I would say that each concept explored a unique approach to positioning each of the sensor components like the PixyCam, the Ultrasonic, and the two IR sensors relative to the main body of the robot and the ball collection system.

The eight concepts that I generated were:

- Elevated head vision - this involved the Pixycam mounted high on a raised angular mount (Inspired by Wall-E), an ultrasonic sensor front and centre, and two lateral (side mounted) IR sensors
- Ground level targeting - here, all the sensors are front mounted at a low height with two IR sensors angled 15 degrees inwards for a smaller detection zone to place the ball right in the middle.
- Dual sensor fusion front plate - here the PixyCam is angled downward, with an ultrasonic sensor right below it, then, it was two IR sensors angled 15° toward the center line for ground level obstacle and ball detection.
- Pan servo camera concept - in this one, the PixyCam is mounted on a metal gear servo with a 120° sweep for a wider field of view, an ultrasonic sensor front centre, and two IR sensors on the corners (oblique). This maximises not only the FOV but also coverage in terms of obstacles.

- Side by side stereo layout - the Pixycam and ultrasonic sensors are mounted side by side here for a compact yet offset design for the face plate. This was made as a layout for the bell enclosure ball collections system.
- Funnel guided intake sensing - the IR sensors are positioned at key trigger zones on the edges of the funnel prongs, the PixyCam is mounted 5° downward statically.
- Tiered sensing zones integrated - this is by far the best design as rated by all team mates. Here, the PixyCam is on a servo (Zone 1 and 2), the ultrasonic sensor (zone 2) relays info to the PixyCam, and the two flank IR sensors are obstacle detectors (zone 2), linked to the locomotion subsystem.
- Minimalist face place - this idea aimed for peak minimalism as I planned an acrylic face plate that houses all sensors statically. This has the same layout as Concept 3 but is fully enclosed in a detachable faceplate with no protruding parts.

See **Figure 11** below for my very own Crazy Eights sensor integration subsystem concept generation.

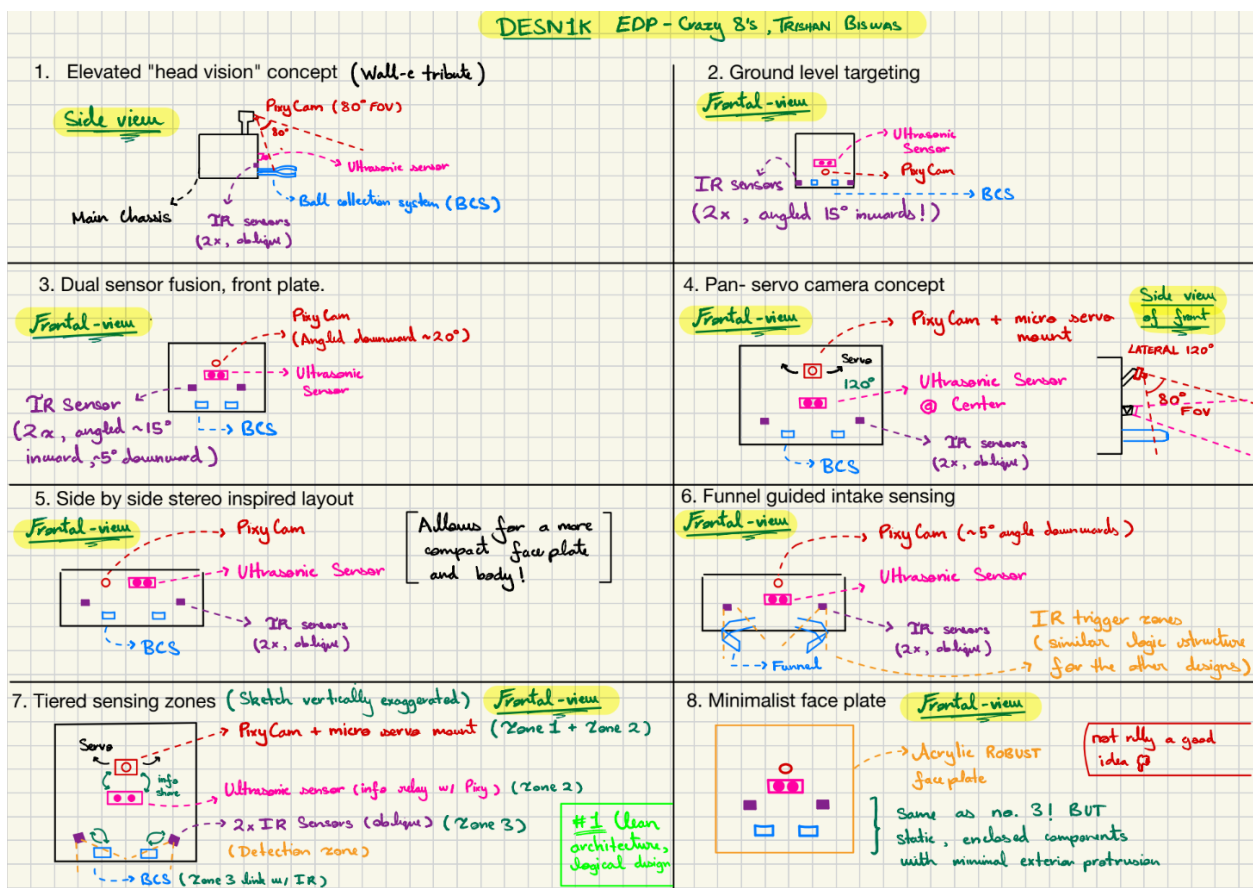


Figure 11: Sensor integration/layout concepts for my crazy eights. Made by me

1.3 Week 3

1.3.1 Entry 7 (02/03)

Entry 7 of the design process section was logged at the start of Week 3. This entry is going to cover how **I carried out** physical sensor and also **evaluated the performance** of the HC-SR04 ultrasonic sensor and the IR obstacle sensor. I also will showcase how I troubleshoot a readout inaccuracy on the ultrasonic sensor through **my coding skills**.

I began the physical sensor tests by first connecting the ultrasonic sensor to the Arduino UNO and wrote a distance measurer code which constantly reads out distance in the serial monitor of the IDE every 300ms. Then, I configured the setup as follows: I placed the red foam ball (given to us) around 15-16cm away from the sensor (measured with a ruler). Subsequently, I ran the program on the IDE, and I read the serial monitor outputs. **At first** everything seemed to be fine as it was outputting values within the range of 15-16cm. **However**, I observed occasional spikes in readings, for an example, see **Figure 12**

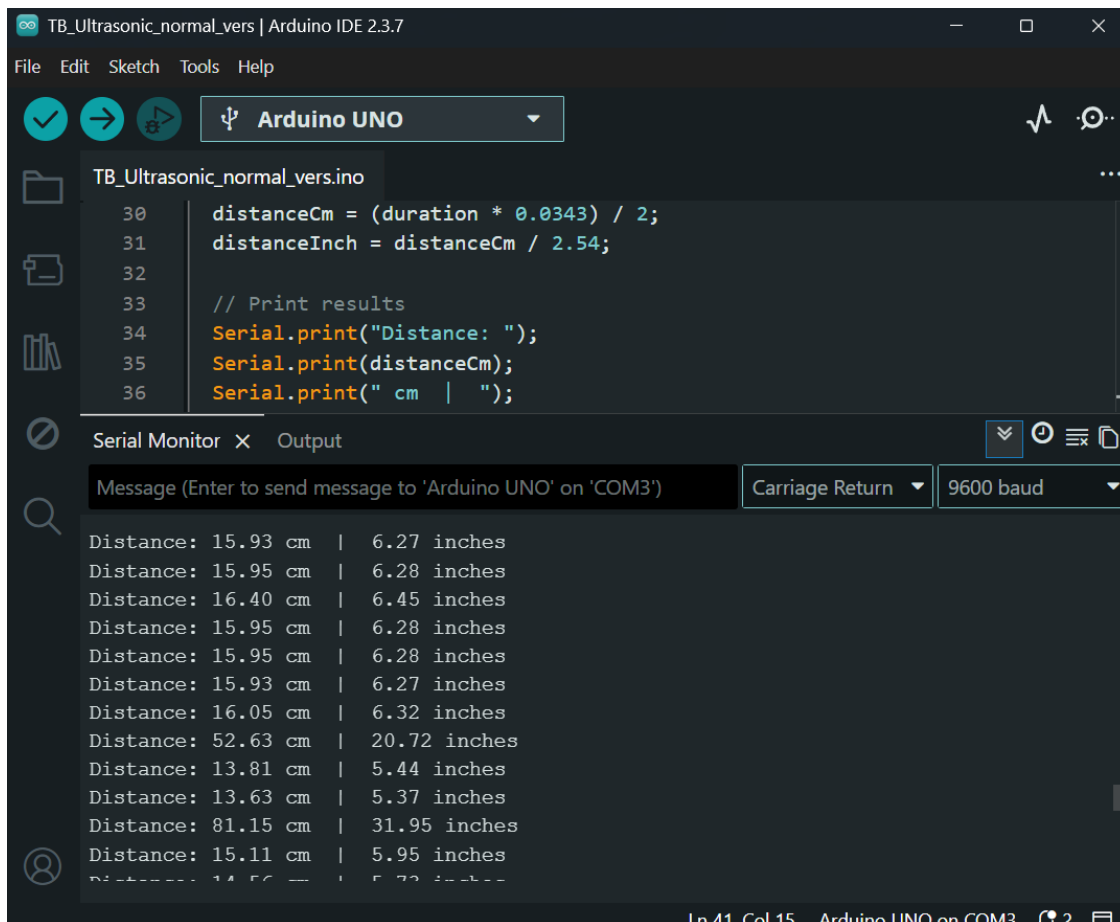


Figure 12: Inaccurate distance readouts due to background clutter with initial code. **Made by me**

I diagnosed this as an issues cause due to signal interference and **NOT** the hardware because it was a source of random error and not systematic. Hence, I amended the code by implementing a “**median filter**” that I made in the code to discard outlier values that spike and

ensure that the logic device gets only the accurate readings. See **Figure 13** for the original code (left) and the median filter version (right).

```

// HC-SR04 Ultrasonic Rangefinder Test
const int trigPin = 9;
const int echoPin = 18;

long duration;
float distanceCm;
float distanceInch;

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.println("Ultrasonic Sensor Test Started");
}

void loop() {
  // Clear the trigger pin
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  // Send a 10us HIGH pulse to trigger
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  // Read echo pin - travel time in microseconds
  duration = pulseIn(echoPin, HIGH);

  // Calculate distance
  distanceCm = (duration * 0.0343) / 2;
  distanceInch = distanceCm / 2.54;

  // Print results
  Serial.print("Distance: ");
  Serial.print(distanceCm);
  Serial.print(" cm | ");
  Serial.print(distanceInch);
  Serial.println(" inches");

  delay(300);
  while(true);
}

```

```

// HC-SR04 Ultrasonic Rangefinder - Median Filter Version
const int trigPin = 9;
const int echoPin = 18;

long duration;
float distanceCm;
float distanceInch;

// ---- Manual Swap Helper ----
void swapValues(long &a, long &b) {
  long temp = a;
  a = b;
  b = temp;
}

// ---- Median Filter Function ----
long getStableReading() {
  long readings[3];
  for (int i = 0; i < 3; i++) {
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    readings[i] = pulseIn(echoPin, HIGH);
    delay(50);
  }

  // Sort and return middle (median) value
  if (readings[0] > readings[1]) swapValues(readings[0], readings[1]);
  if (readings[1] > readings[2]) swapValues(readings[1], readings[2]);
  if (readings[0] > readings[1]) swapValues(readings[0], readings[1]);
  return readings[1];
}

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  Serial.println("Ultrasonic Sensor Test Started (Median Filter Active)");
}

void loop() {
  // Get stable median reading
  duration = getStableReading();

  // Calculate distance
  distanceCm = (duration * 0.0343) / 2;
  distanceInch = distanceCm / 2.54;

  // Print results
  Serial.print("Distance: ");
  Serial.print(distanceCm);
  Serial.print(" cm | ");
  Serial.print(distanceInch);
  Serial.println(" inches");

  delay(300);
  while(true);
}

```

Figure 13: Code for ultrasonic distance measurement (read line above figure). **Made by me**

After applying the median filter, the readings were far more accurate and also consistent. This confirmed that the sensor was actually reliable in the use case that I'm planning to use this for - for approach distance measurement to communicate to the Arduino and relay it to the motor driver. See **Figure 14** below for just the median filter amendment I applied and continue reading for an explanation of how it works.


```

median filter extract

1  // ---- Manual Swap Helper ----
2  void swapValues(long &a, long &b) {
3      long temp = a;
4      a = b;
5      b = temp;
6  }
7
8  // ---- Median Filter Function ----
9  long getStableReading() {
10     long readings[3];
11     for (int i = 0; i < 3; i++) {
12         digitalWrite(trigPin, LOW);
13         delayMicroseconds(2);
14         digitalWrite(trigPin, HIGH);
15         delayMicroseconds(10);
16         digitalWrite(trigPin, LOW);
17         readings[i] = pulseIn(echoPin, HIGH);
18         delay(50);
19     }
20     // Sort and return middle (median) value
21     if (readings[0] > readings[1]) swapValues(readings[0], readings[1]);
22     if (readings[1] > readings[2]) swapValues(readings[1], readings[2]);
23     if (readings[0] > readings[1]) swapValues(readings[0], readings[1]);
24     return readings[1];
25 }
26
27 // Called in loop() instead of a single read
28 duration = getStableReading();
29
30

```

Figure 14: Median filter code (just the filter part extracted). **Made by me**

What the filter does essentially is that it takes 3 consecutive readings. So instead of firing it once, the “**getStableReading()**” fires a distance measurement three times. Then “**swapValues()**” does a simple bubble sort on the array of three values received and only **then**, the middle (median) value is returned “**readings[1]**.” Furthermore, any “spike” in the values sits at the bottom of the array and is discarded. Furthermore, see **Figure 15** for an image of the Ultrasonic sensor test video I shared on Teams.

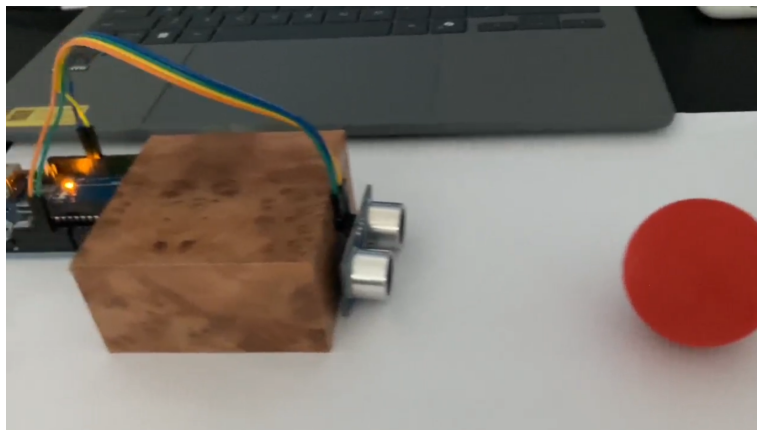


Figure 15: Picture of ultrasonic test video shared on Teams. **Made by me**

Next, the IR Sensor testing. This was quite straightforward as the IR sensor module is quite easy to set up with the Arduino and to code on the to output "No object detected" or "Object detected" on the Serial Monitor. I knew the IR sensor was working well not only due to the fact that the serial monitor was reading out "Object detected," but rather because the LED on the IR module turns on. I also tweaked the manual distance screw on the module to test it at a slightly longer distance. The code is as follows in **Figure 16** below for the IR sensor and the picture of the video that I shared on teams follows that Figure in **Figure 17**.

```
DESN1000 · TEAM C01 · SENSOR TESTING
IR Obstacle Sensor — Proximity Detection
Digital detection via built-in LED and manual range calibration screw

TB_IR_proximity_closerange.ino

1 // IR Obstacle Sensor Test
2 const int irPin = 7;
3 int sensorValue;
4
5 void setup() {
6   Serial.begin(9600);
7   pinMode(irPin, INPUT);
8   Serial.println("IR Sensor Test Started");
9 }
10
11 void loop() {
12   sensorValue = digitalRead(irPin);
13
14   if (sensorValue == LOW) {
15     Serial.println("Object DETECTED");
16   } else {
17     Serial.println("No object detected");
18   }
19
20   delay(200);
21 }
```

Figure 16: Code for the IR sensor, LOW/HIGH is detected/undetected. Made by me

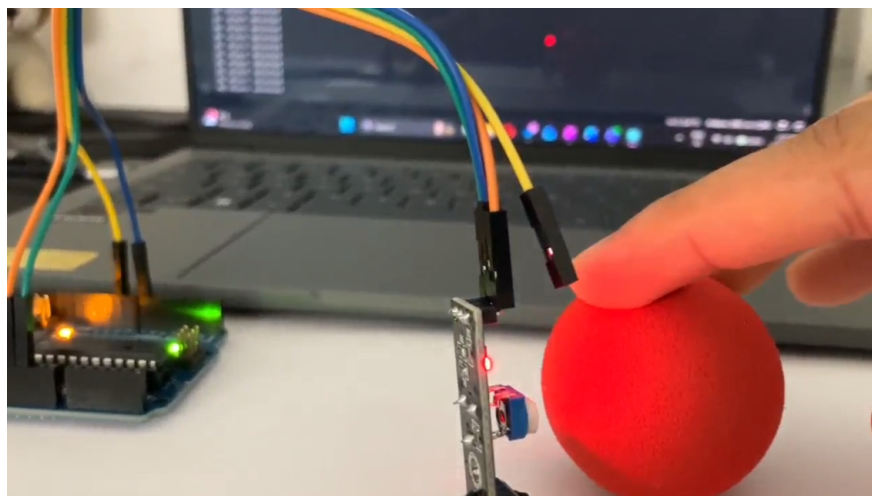


Figure 17: Picture of ultrasonic test vid I shared on Teams. Made by me

1.3.2 Entry 8 (05/03)

Entry 8 will purely consist of EDP work. As that is the main priority due to this assessment task being weighted 20%. Hence, for this entry I will be discussing the progression of our problem statement and also our team's LucidSpark mindmap and the morphological chart generated by Franklin for our three chosen concept generation tools.

The problem statement was first brainstormed as a team based on information from Wendy's W2 lecture. Subsequently, we continually edited it as a team by providing input. And finally, once we prepared it enough, we went to Brielle and Wendy to receive feedback, upon receiving the feedback. Jesse and Yusuf incorporated them into a final problem statement that is very solid, well defined, and specific. Here is the progression of it:

“Design and construct an autonomous robot that identifies and sorts plastic waste into PET, HDPE and PP categories while avoiding non-recyclable materials. The primary stakeholders are the UNSW Environmental Sustainability Team, along with students and staff, who require more accurate and efficient sorting of plastic waste to prevent contamination of recycling streams and improve recycling effectiveness. The robot must be a single, self-contained unit with dimensions no greater than 200 mm × 200 mm × 200 mm

→ Redrafted as a team

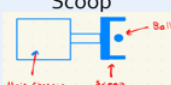
Design and build an autonomous robot that navigates the arena, identifies and collects the operator-specified recyclable plastic (red = HDPE, green = PET, blue = PP), avoids yellow (non-recyclable) items and obstacles, and returns the collected item to the team base. The primary stakeholders are the students and staff, who require this robot to remove waste across campus and improve recycling accuracy. The robot will also operate as a single and self-contained physical unit, start within the designated starting zone, have initial dimensions no greater than 200 mm × 200 mm × 200 mm, and use no more than \$50 of additional materials beyond the provided kit.”

→ Final one

Design and construct a prototype of an autonomous robot that identifies and sorts plastic waste into PET, HDPE and PP categories while avoiding non-recyclable materials. The primary stakeholders are the UNSW Environmental Sustainability Team, along with students and staff, who require a more accurate and efficient sorting solution over manual sorting, to reduce contamination of recycling streams and improve recycling effectiveness. The robot must be a single, self-contained unit with dimensions no greater than 200 mm × 200 mm × 200 mm.

These were achieved using the 5Ws scaffold, by following the marking rubric closely and identifying things that are missing or need rephrasing on our own. Followed by, a final presentable version which was informed by clear feedback from Brielle and Wendy in response to our prepared questions for them.

Now, for the concept generation process, here are the morphological chart (designed by Franklin) and the LucidSpark mindmap that we all worked on, followed by Bradman, Yusuf and Yizhuo's drafting of the slides. See **Figures 18** and **19** for the two other concept generation tools used in this EDP preparation process.

Means	1	2	3	4
Functions				
Explore arena	Tank tracks 	3 wheels 	2 wheels and caster 	4 wheels 
Detect ball and obstacles	Parking sensors 	Computer vision 	Pressure sensor 	Ultrasonic sensor 
Identify colour	Camera 	Colour sensor 	Spectral sensor 	
Collect and release ball	Clamp 	Bell 	Roller 	Scoop 
Navigate to correct bin	Line markings 	Computer Vision 	Remote control 	

TEAM C01 · DESN1000 · Engineering Design Process

22 / 28

Figure 18: Morphological chart designed by **Franklin**.



TEAM C01 · DESN1000 · Engineering Design Process

21 / 28

Figure 19: The LucidSpark mindmap created by the team as a whole.

1.3.3 Entry 9 (07/03)

Based off of our previous meeting, we established that all team members will work independently over the weekend on their EDP presentation sections assigned to them. For me, that was the concept solutions section. That meant, I had quite a bit to get done. Firstly, I convened with the subgroup leaders on Friday to narrow down concept solutions based off of the discussions we had in relation to our concept generation tools as a team in our last meeting. We met via Teams on multiple days across the weekend. And these following base concept solutions are what we concluded:

- Concept 1: a robot with a passive funnel scoop with an enclosure to store the balls. It's sensor layout would be static mounted but angled appropriately with a high mounted PixyCam, one middle ultrasonic sensor, and two lateral mounted IR sensors. The wheels would be triangular tank tracks (like Wall-E).
- Concept 2: a robot with a vertical "bell" enclosure which descends upon the ball (actuated by DC motor) and securely holds it. With a 4 wheel, 2 rear caster wheel configuration with FWD. And the sensor layout would be angled and mounted in an offset manner to account for the ball collection system in the middle with no ultrasonic sensor and two front IR sensors for ball positioning.
- Concept 3: a servo aided and rubber band/rubber lined clamp design to grip the ball securely. It would have a 4 (regular) wheel layout with FWD. And the sensor layout would be a PixyCam mounted high angled downward on a sweeping servo to maximise FOV and two ultrasonic sensors front centre for ball and ball cluster rangefinding along with two flank IR sensors. Additionally for aesthetics, a Wall-E head will be printed along with a Wall-E chassis (minus the tank tracks).

Now, the next steps for me is to sketch them with annotations and labelling. I will begin sketching the concepts on Sunday and aim to have them finished by Monday's team meeting. After this meet, I began working on 2 minute long **my EDPP script**, I drafted and **asked team members** for feedback and eventually produced a final version shown in **Figure 20**

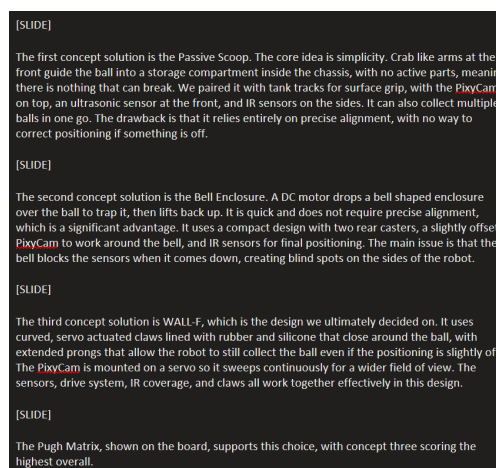


Figure 20: My final EDPP script uploaded to team OneNote for the concept solutions section.

1.4 Week 4

1.4.1 Entry 10 (09/03)

I had finally finished the concept solution sketches with the annotations for each of the three concepts. See **Figure 21**, **22**, and **23** for concept solutions 1, 2, and 3.

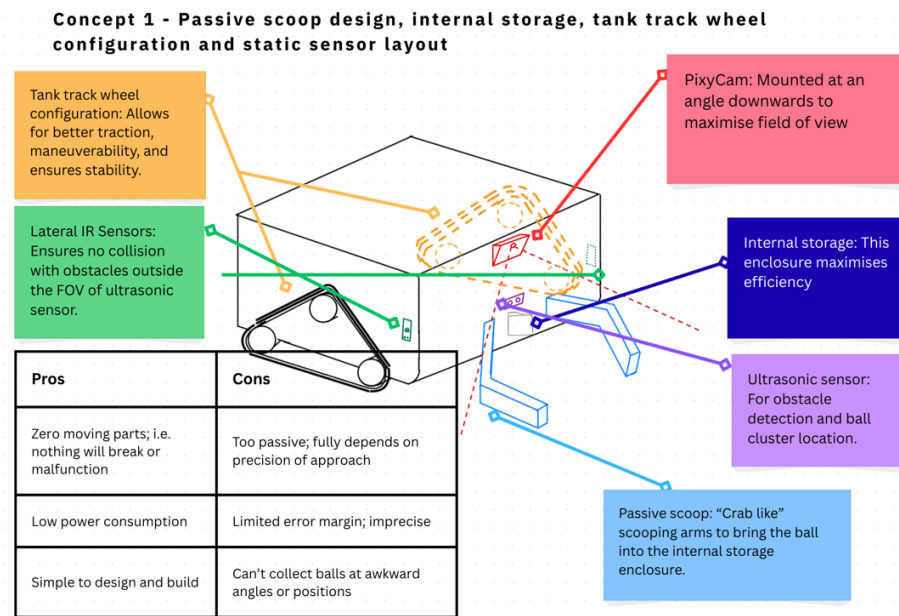


Figure 21: Concept solution 1: Passive funnel scoop prototype sketch. Made by me

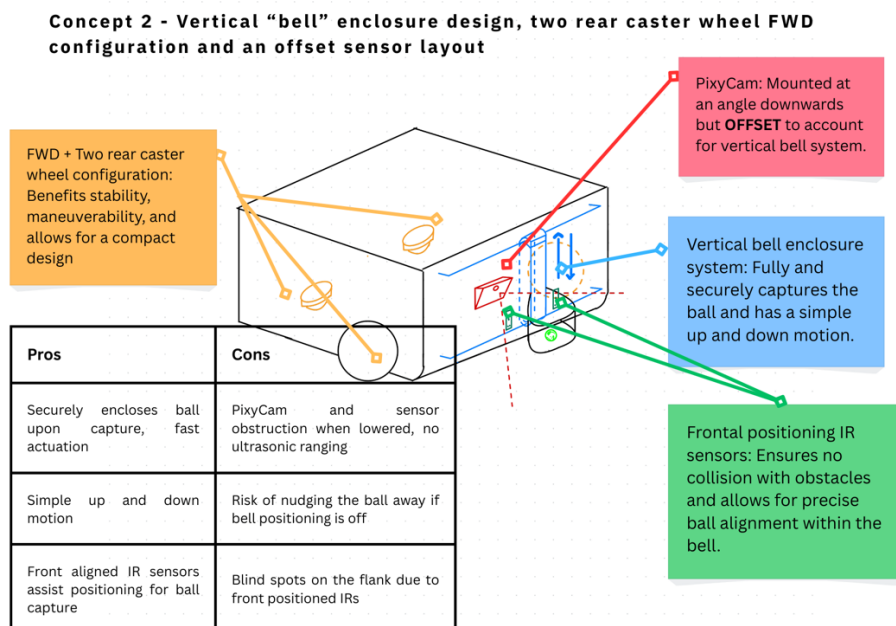


Figure 22: Concept solution 2: Vertical bell enclosure prototype sketch. Made by me

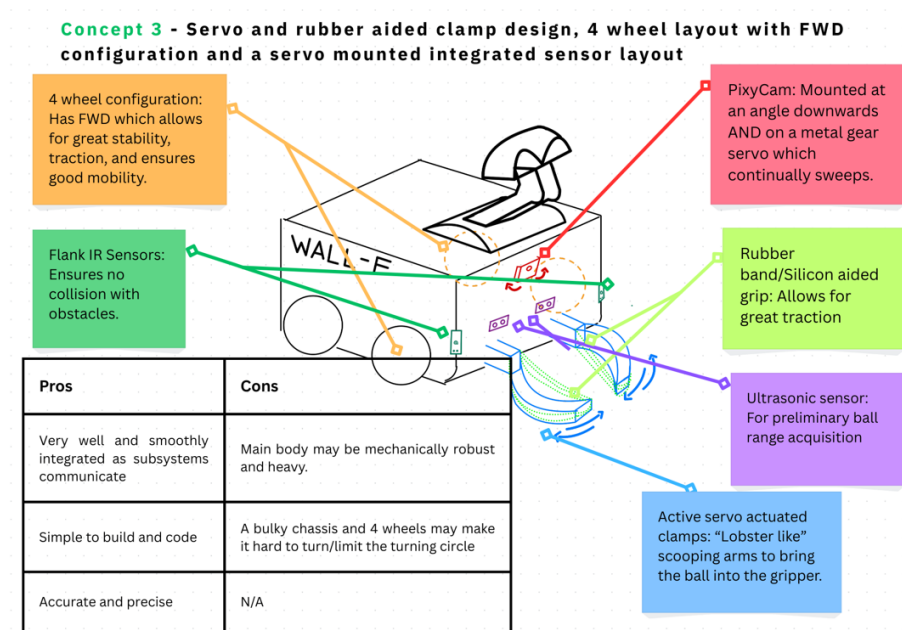


Figure 23: Concept solution 3: Servo aided clamp prototype sketch. **Made by me**

Having completed all the concept sketches with appropriate labelling and annotations, I have met my goal established from the last entry and I am right on schedule for the team set EDP deadlines. This meant that I could show up to our Monday group meeting with solidified concept solutions sketched out for an evaluation by the whole team. This process involved the whole team utilising the “Pugh Matrix,” which is a ranking system that involves settings a “baseline” or a standard, and whether each concept has a characteristic above or below the required standard. **I took initiative** and created the Pugh matrix before hand. Having done through this as a whole team, the results concluded that Concept 3 is the most mutually favoured solution to go ahead with. See **Figure 24**.

Pugh Matrix — Concept Evaluation

DESN1000 · Team C01 · ELEC - Waste Sorting Robot

Made by Trishan Biswas

Evaluation Criterion	C1 — Passive Funnel	C2 — Bell Enclosure	C3 — Servo Gripper ★
Ball collection reliability	0	-	+
Mechanical complexity	+	-	0
Sensor integration	0	0	+
Fabrication feasibility	+	-	+
Positioning tolerance	-	-	+
Power efficiency	+	0	0
Total Score	+2	-3	+3

Key

+ Better than baseline 0 Same as baseline - Worse than baseline
★ Selected concept

Figure 24: The Pugh matrix after the team and I evaluated the concepts. **Made by me**

1.4.2 Entry 11 (12/03)

For entry 11, which is based off of 12th March's group meeting where we continued our EDP presentation scripts and speaking it individually and timing it for further script refinement. We also aimed to lock everything in within this group meeting such as the script, make a cohesive script which combines everyone's scripts and we add stage directions, transitional sentences and practice our delivery very roughly before a proper and complete rehearsal on Saturday and Sunday. Furthermore, we planned to thoroughly prepare every single member on slides other than theirs over the weekend in preparation for the Q&A segment where anyone can be asked on anything in the presentation.

See **Figure 25** below for a portion of the complete EDP Presentation script that we all collaborated on and contributed to (but mainly Jesse).

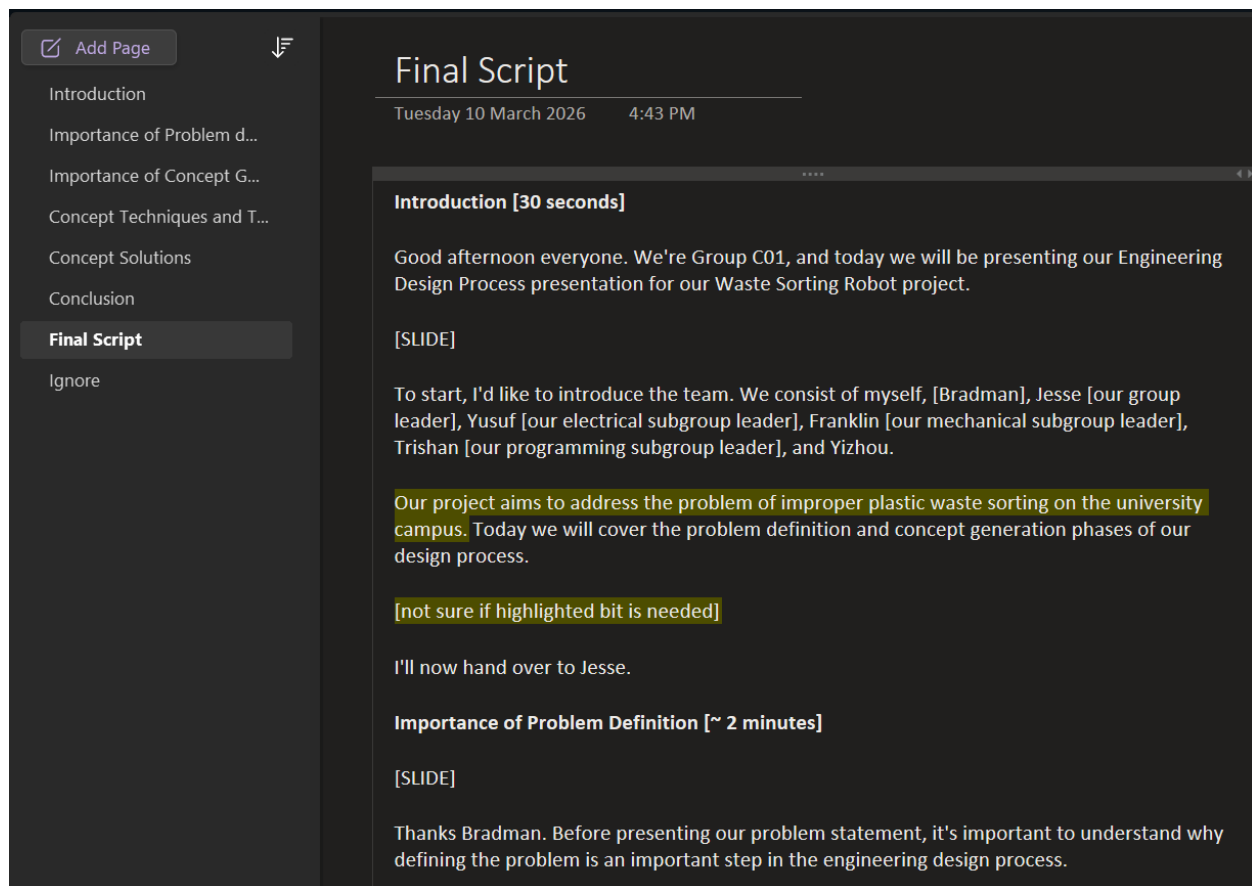


Figure 25: A portion of the final script. With transitional sentences, stage directions, and each section's script.

Furthermore, in terms of the refinement of the EDP slides, it was also done during the session, led by Bradman. We originally went from a dark themed aesthetic which he deemed as unclear in terms of readability. Thus, we switched to a calmer and more legible white colour scheme and increased the font sizes of the text. Furthermore, we strictly aimed to keep text absolutely minimal but concise.

2 Teamwork

2.1 Week 1

2.1.1 Entry 1 (18/02)

As it is Week 1, in terms of teamwork, we conducted an informal team meeting on the 18th of February where we discussed everyone's strengths and areas of comfort regarding the key components of this project (i.e. coding, mechanical, and electronics). As of now there have been no formal role assignments, as we are waiting for our first mentor meeting. Furthermore, in the informal meeting we established that we are all aiming for a High Distinction. And, that we all seem to be on the same page in terms of course outcomes and the direction that the project will be heading. Concluding the meeting, there is a general sense of subgroups, however, we have not finalised a leader yet.

Time	18/02, 13:00-14:00
Location	[In person UNSW Main Library]
Attendance	All members were present.
Agenda	• Discuss ball retrieval mechanism options • Explore wheel configurations and layout • Discuss team strengths and comfort areas and evaluate possible subgroup leaders • Gain a general idea on project and course outcomes and the effort willing to be put in
Meeting Summary	• Preliminary clamp/gripper designs discussed and evaluated • Wheel configurations compared (3-wheel vs 4-wheel) and layout options (caster rear wheels or normal rear wheels) • Informal discussion of individual strengths in conjunction with establishing that we all are aiming for a HD in this course and project.
Decisions Made	• Claw/gripper mechanism identified as preferred retrieval approach • 3-wheel and 4-wheel configurations both still under consideration • Formal role assignments deferred to next session after mentor meeting

Action Items	• All members will reflect on preferred roles prior to next meeting • Trishan will look into PixyCam colour detection capabilities before further sensor work and planning. • Team to review and walk through Design Journal 1 guidelines • Team will aim to complete Makerspace Induction and Soldering badge before/during the day of first mentor meeting.
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2.1.2 Entry 2 (20/02)

Time	19/02, 15:00 – 17:00
Location	ElecEng224
Attendance	All members + Mentor were present.
Agenda	• Determine team roles and jobs for the weekend • Establish recurring meeting times • Acquire Soldering Badge
Meeting Summary	• Team roles formally established across four subgroups: Documentation, Electrical, Programming, Mechanical/Chassis • Weekly internal meetings set for Mondays at 12:00pm in the library • All members obtained Soldering Badge
Decisions Made	• Weekly internal meetings on Mondays at 12:00pm in the library; additional meetings online as needed • Roles established across Documentation, Electrical, Programming, and Mechanical/Chassis subgroups

Action Items	• Trishan to research programming with all sensors and set up a component layout • Franklin to measure Arduino casing dimensions and plan the Arduino UNO layout with chassis • Bradman to refine preliminary ball collection system designs. • Jesse to research and compare wheel design options • Yusuf awaiting more information on the electrical component of the design but continuing individual contributions to the project. • Yizhuo to be in charge of documentation and research.
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Significant progress was made in our very first mentor meeting as subgroups, team leader, and subgroup leaders were selected and finalised early on. This was very good as it gave everyone a sense of direction early, which is crucial to the smooth development of our robot. See the table below for the **team roles and responsibilities** assigned.

Role	Responsibilities	People
Documentation	• Maintain notebook • Record meeting notes • Track design changes	• Jesse • Yizhuo • Bradman
Electrical	• Wiring and circuit layout • Sensor integration • Troubleshoot electrical issues	• Yusuf (Subgroup Leader) • Franklin • Trishan
Programming	• Write and test code • Ensure accurate autonomous operation • Ensure seamless integration of all subsystems • Debug software issues	• Trishan (Subgroup Leader) • Franklin • Yusuf

Mechanical & Chassis Design	• Design robot structure • CAD modelling • Assembly and fabrication • Mechanical testing and adjustments	• Franklin (Sub-group Leader) • Bradman • Jesse • Yizhuo
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2.1.3 Entry 3 (21/02)

Time	20th-21st Feb
Location	Informal Conversations / Independent
Attendance	All members contributing independently
Agenda	• Individual progress on assigned tasks • Team updates via Microsoft Teams
Meeting Summary	• Trishan (me) researching kit items and components, and their roles. Furthermore, creating the LucidSpark sub-system map. • No formal meeting held, progress tracked through Teams announcements and informal check ins and conversations • Franklin progressing on clamp design refinement and Arduino Casing on Fusion • Bradman refining wheel types, clamp and functionality options • Jesse contributing to documentation and leading weekly comms, due to the leader role.
Decisions Made	• Component roles and subsystem assignments finalised • LucidSpark adopted as shared visual tool for system mapping

Action Items	• Trishan to begin sensor programming research ahead of Week 2 • Franklin to use component layout to inform chassis dimension planning • Bradman to finalise wheel config recommendation for next team meeting
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2.2 Week 2

2.2.1 Entry 4 (23/02)

Time	23/02, 12:00 – 13:00 and 15:00 – 17:00
Location	Main Library and ElecEng224
Attendance	All members were present.
Agenda	• Review prelab content and share previous week's progress • Go over EDP requirements as a group • Establish team workflow, deadlines, and communication structure
Meeting Summary	• 4 wheel configuration formally selected for superior stability • Claw clamp selected as preferred ball collection mechanism • EDP structure reviewed: problem definition, conceptual design importance, annotated sketches • Discussed HD requirements for EDP and formulated questions for mentor • Trishan troubleshoot and resolved PixyCam compatibility issues on macOS • Franklin designed Arduino mounting plates and smaller component mounts • Bradman created EDP PowerPoint scaffolding

Decisions Made	• 4 wheel configuration selected as final locomotion design • Claw clamp selected as ball collection mechanism • Goal oriented subgroup workflow with end of week synthesis and constant check ins • Deadline structure: physical tasks 1 week ahead, writing 3 to 4 days ahead, hard deadline 1 day before submission • Work ahead of time principle adopted to prevent time crunch • Jesse to run weekly workload questionnaire flagging if members have too much or too little on • Progress tracking via Excel and Gantt chart • EDP slides to address marking criteria directly; verbal delivery can be more flexible
Action Items	• Begin wheel fabrication early • Finalise component mounting positions • Get Arduino operational • Begin EDP presentation, address marking criteria directly in slides

2.2.2 Entry 5 (26/02)

Time	26/02 12:00 – 13:45
Location	Main Library and ElecEng224
Attendance	All members were present.
Agenda	• Draft problem statement using lecture slides and recording from Wendy's W2 lecture • Discuss and narrow down 3 concept generation tools we're going to use • Plan important questions for Brielle and possibly Wendy regarding problem statement • Complete, agree upon, and discuss TEAM CONTRACT with Brielle

Meeting Summary	• We used the 5Ws framework to scaffold the problem statement, then filled in details including key constraints • We then planned to seek feedback from our mentor and Wendy directly on the current problem statement, regarding what is working and what is missing from it. • Then, we all discussed and narrowed down concept generation tools to a morphological chart, LucidSpark mindmap, and Crazy Eights. • Then, the crazy eights task was assigned per subsystem: Trishan (sensor integration), Bradman (wheel design), Jesse and Franklin (gripper design), Yusuf and Yizhuo (aesthetics) • We walked through the team contract in detail, establishing guidelines and safety nets in case of worst case scenarios.
Decisions Made	• By the end, morphological chart, LucidSpark mindmap, and Crazy Eights confirmed as concept generation tools • The 5Ws adopted as the framework for problem statement formulation along with identifying constraints and following the marking criteria. • We agreed to refer to the team contract in the event of worst case scenarios or to resolve disputes. And we said that we'd all adhere to the expectations we set on it with Brielle
Action Items	• After today, each member must complete their own Crazy Eights for their assigned subsystem by the 1st of March. • Seek feedback from mentor and Wendy on problem statement before finalising and incorporate the feedback into the problem statement. • We must begin populating morphological chart and mindmap and have it completed by Week 3 Monday.

2.2.3 Entry 6 (27/02)

Time	Nearing the end of W2
Location	Independent / Microsoft Teams
Attendance	All members contributing independently
Agenda	• Firstly, we all will review each member's contribution to their assigned subsystem's Crazy Eights. • Then, to continue contributing individual subsystem ideas to a shared LucidSpark mindmap that I created and shared. • Franklin will independently make a morphological chart that we will all work on to decide concept solutions.
Meeting Summary	• (It was very good to see that) All members were working on their Crazy Eights and doing multiple trials to refine it. • Trishan – sensor integration • Bradman – wheel design • Jesse and Franklin – clamp/gripper design • Yusuf and Yizhuo – aesthetics • Jesse noted that all members contributed to the LucidSpark mindmap for their respective subsystems by the set deadline (team set, not the formal deadline) • Furthermore, Franklin produced the morphological chart covering sensors, wheels, clamp type, and navigation. • Bradman collated these ideas and put them on the EDP PowerPoint presentation

Decisions Made	• Crazy Eights, LucidSpark mindmap, and morphological chart confirmed as the three concept generation tools for the EDP presentation. • We narrowed down that the 4 best crazy eights design will be going on the presentation. • We all decided that Bradman would be in charge of the PowerPoint presentation and updating it (along with Jesse). • While we get more info regarding the EDP, Trishan will be conducting sensor tests in Week 3 soon. • Franklin will be continuing on the mechanical side of things leading the subgroup towards wheel design.
Action Items	• Complete the MindMap before W3 Tuesday at the latest, everyone must continue contributing to it. • Hold an informal meeting with the whole team to review the Morph chart and shortlist the best ideas. • Eventually hold another informal between subgroup leaders to finalise concept solutions based on the MindMap, Crazy eights, and Morph Chart. With, a final Pugh Matrix ranking of the three draft solutions to shortlist the FINAL chosen concept solution.

2.3 Week 3

2.3.1 Entry 7 (02/03)

Time	02/03, 12:00 – 13:45
Location	Main Library
Attendance	All members were present.
Agenda	• We should briefly review each other's Crazy Eights submissions • Main agenda is to refine the problem statement as a group and draft/re-draft multiple times • We must prepare good questions for Brielle/Wendy regarding the Problem Statement

Meeting Summary	• All of the team members reviewed and discussed each other's Crazy Eights across all the subsystems. And we shortlisted 4 of the best. • Then, the problem statement was edited multiple times with input from all members to produce a refined version ready for feedback from Brielle or Wendy • We prepared questions for Brielle and Wendy regarding the problem statement • Then we went over the LucidSpark mindmap we completed as a team and we agreed that images need to be added for clarity. So it is not yet fully complete.
Decisions Made	• Problem statement version prepared to present to Brielle and Wendy for feedback • We planned that some images must be added to LucidSpark mindmap for improved clarity by Wednesday at the latest. • The best Crazy Eights Designs that were chosen were Franklin, Jesse, Yusuf and Trishan's designs that are going on the presentation.
Action Items	• We should finalise and lock in the problem statement by Thursday after feedback. • We must review the morph chart and shortlist best ideas across subsystems as a team (if not subgroup leaders) • We should aim to have three concept solutions defined by end of Week 3 or start of Week 4

Time	02/03, 16:00 – 17:00
Location	ElecEng224
Attendance	All members were present.
Agenda	• We must present the current problem statement draft for feedback • Ask prepared questions regarding problem statement content and structure

Meeting Summary	• We presented refined problem statement to Brielle and Wendy and received feedback on what to add or adjust. • From Brielle, we need to remove any constraints within the problem statements that are not necessary. • From Wendy, we need to add a “when”. As in, when is the project required to be completed by? And further clarity is required on the “who” part. As in, who are the specific stakeholders/consumers at play.
Decisions Made	• Thus, we aimed to have the problem statement updated, based on the feedback by Thursday this week.
Action Items	• Jesse and Yusuf to incorporate Brielle and Wendy’s feedback into the final problem statement, and have a proper “final” one ready. • We should all proceed with concept solution development before the end of Week 3 in order to have 3 fully fixed concept solutions by W4.

2.3.2 Entry 8 (05/03)

Time	05/03, 12:00 – 13:45
Location	Main Library
Attendance	All members were present.
Agenda	• We all walk through and discuss the morph chart, mindmap, and (briefly) the Crazy Eights as a team for shortlisting concept solutions. • We make enough notes and suggestions for subgroup leaders to generate three distinct concept solutions. • We must assign ownership of concept finalisation and sketching for the EDP presentation • Read through our locked in problem statement. • IMPORTANT: to assign every member a specific section of the EDP for them to work on over the weekend.

Meeting Summary	• The team went through the morph chart and the LucidSpark mindmap together, along with a brief revisit and discussion of the Crazy Eights designs. • Enough input was given by everyone for subgroup leaders to take to narrow down three concept solutions and eventually refine it to one. • Trishan has been assigned to sketch, annotate, and label all three concepts for the EDP presentation slides. • Over the weekend, subgroup leaders must shortlist the concept solutions by end of Week 3 • The whole team must complete/ have some verbal input into the Pugh Matrix by start of Week 4. • End of Week 3 is reserved for individual work on everyone's EDPP scripts
Decisions Made	• As discussed above. And for the EDP roles are: • Jesse and Yusuf – Importance of problem statement • Importance of concept generation (technical and financial risk) – Yizhuo • Importance of concept generation (Innovation and performance) – Bradman • Concept generation tools – Franklin • Concept solutions – Trishan
Action Items	• Franklin, Trishan, and Yusuf to finalise the three concepts concretely. • Trishan must make annotated concept sketches for the EDP slides • All subgroup leaders will shortlist the best ideas over the weekend via Teams/informally. • Everyone HAS TO begin drafting individual EDP scripts by end of Week 3 and finalise by Wednesday Week 4.

2.3.3 Entry 9 (07/03)

Time	06/03 – 08/03
Location	Independent work / Microsoft Teams
Attendance	All members contributing independently
Agenda	• Each member continuing work on their individual EDP presentation section to completion and prepare script. • Subgroup leaders will shortlist and finalise three base concept solutions • Trishan to begin concept sketches, labelling, annotation, and Pugh Matrix
Meeting Summary	• No formal meeting was held as progress is being co-ordinated informally via OneNote and the PowerPoint across the weekend • Subgroup leaders informally met via Teams to shortlist three final concept solutions from the team discussion and key inputs from members defined on 05/03 • Trishan began sketching all three concept solutions for the EDP slides. The annotations and labelling are supposed to be completed by Week 4 Monday. • All members worked independently on their respective EDP sections.
Decisions Made	• The three final concept solutions shortlisted and confirmed by subgroup leaders
Action Items	• Trishan will complete concept sketch and the respective annotations by Week 4 Monday • All members will continue refining their individual EDP sections and creating a script to memorise and present to the team • The whole team will start the Pugh Matrix together in the Week 4 meeting.

2.4 Week 4

2.4.1 Entry 10 (09/03)

Time	09/03, 12:00 – 13:45
Location	Comp Lab on Level 2
Attendance	All members were present.
Agenda	• We have to review each other's EDP presentations, work, and scripts and brief the team on each other's sections briefly before a full briefing later on in W4. • Main agenda is to complete the Pugh Matrix as a team • Based on results, we have to select a final concept solution to go ahead with. • Continue with script memorisation and streamlining the presentation and its coherence (Bradman and Jesse in charge for this).
Meeting Summary	• In this meet, each member briefly presented their EDP section to the rest of the team for a brief review and feedback session. • Trishan presented the completed concept sketches with full annotations briefly. • The entire team evaluated all three concept solutions together and informally completed the Pugh Matrix. • Concept 3 (the Servo gripper or “WALL-F” prototype) was formally selected as the final design based on the Pugh Matrix outcome. • The Pugh Matrix was updated and prepatated to go in the EDP PowerPoint presentation. • The next steps for each member's EDP section was discussed. Mostly just memorising, presenting and timing

Decisions Made	• Concept 3 selected as the final robot design because it scored the highest across the various criteria in the Pugh Matrix • Bradman and Jesse in charge to refine the PowerPoint and make it look good and coherent • Trishan in charge of timing when we run through a draft presentation. • Yizhuo and Bradman in charge of introduction and conclusion • Yusuf in charge of sources/bibliography
Action Items	• Everyone must continue refining and rehearsing their individual EDP sections in their own time. We should all time it too. • We should finalise the EDP PowerPoint and make it ready for submission by the end of Saturday, W4 (One day before deadline).

2.4.2 Entry 11 (12/03)

Time	12/03, 12:00 – 13:45
Location	Computer Lab
Attendance	All members were present.
Agenda	• We reviewed and trimmed each member's individual script by asking them the timing on their speech and modifying the script accordingly • We all collaborated to write transitional sentences between sections, led by Jesse • We then combined all scripts into one cohesive master script with stage directions. • We all worked towards finalising and refining the EDP slides. We aimed to ensure that it is ready for a full submission after Saturday's morning rehearsal. Led by Bradman

Meeting Summary	• We first trimmed each member's individual script and reviewed it collaboratively to improve conciseness and the overall flow of the master script • Jesse led writing transitional sentences for each section to handover to the other. We wanted to ensure that the delivery was cohesive across all speakers. • By the end of the meeting, a master script was produced which combined all sections with slide change directions and speaker cues. • We gave some special attention given to Yizhuo and Bradman's intro and conclusion sections to ensure that there is a smooth opening and closing. • Finally we reviewed the EDP slides and finalised the aesthetics. This was mainly led by Bradman.
Decisions Made	• The master script has been finalised and confirmed as everyone's point of reference to practice the presentation and look over others' sections • The EDP presentation was locked in following final review by the team leader Jesse on Saturday.
Action Items	• We decided that all members must rehearse their individual sections independently before Saturday where we will have a morning rehearsal session and an in person session on Sunday. Following us coming to the campus early on Monday well before our EDP presentation time slot to rehearse everything one last time. • We should all memorise key points to avoid going on tangents and off script.

3 Reflections

3.1 Week 1

3.1.1 Entry 1 (18/02)

What am I doing well?

- I took initiative alongside the team in raising the importance of clamp designs and wheel configurations early on in the project as it would influence decisions later on.

Furthermore, as a team we evaluated that I'm the most comfortable with the electronics and coding side of things, thus, I took the lead and initiative in running a preliminary test with the PixyCam. That is, colour detection.

What is our team doing well?

- There was very good engagement and enthusiasm in the first informal session as everybody contributed to the discussion and provided their own input. And everyone was on the same page of aiming for a HD in the course and project.

Why were the task(s) important?

- The wheel configuration task was crucial as it determines the chassis design and manoeuvrability. Therefore, getting this right early would mean future redesigns and issues will be mitigated.
- The clamp design is also central to the robot's functions, as establishing the roadmap now gives the mechanical subgroup a concrete starting point and successfully addresses the criteria.

What could I have/can I do better?

- I should have documented the sketches with annotations and labels during the session rather than after it.
- I could have pushed myself and my probably subgroup members for some preliminary component role assignments rather than leaving it for the next session.
- I should have requested Bradman to construct a more formal sketch.

Is the progress in alignment with schedule?

- Yes it is, as we made the most out of week 1 by focusing on familiarisation and early preliminary conceptualisation, and no project deliverables were missed.

What have I learned? What do I need to learn?

- I learned the pros and cons of 3 wheel and 4 wheel configurations. I need to learn the code/torque specifications for the servo to validate the functioning of the claw. I communicated early on the importance of deciding wheel config as early as possible. Because it cascades into further mechanical development.

3.1.2 Entry 2 (20/02)

What am I doing well?

- I took ownership of setting up the PixyCam independently, troubleshoot driver issues and resolved them, and I took initiative in setting up colour signature detection early through my self-directed problem solving and testing.

What is our team doing well?

- That we formalised roles early to give everyone clear direction, responsibilities and accountability going into the actual designing/build phase.

Why were the task(s) important?

- This task was vital as the PixyCam is the main sensing component; which determines whether the right waste is being sorted or not. Without it, the robot cannot fulfil its core purpose. Moreover, formalising roles prevents duplication of work and effort while also ensuring that all subsystems have their own dedicated team.

What could I have done better?

- I should have researched/watched a video of PixyCam setup prior to wasting 1 hour of my time on troubleshooting to save time. I could've documented the exact saturation for later testing and reproduction.

Is the progress in alignment with schedule?

- Yes, the roles are assigned early and the first preliminary sensor test are completed within Week 1. This is keeping us on track for further tests, the EDP presentation, and the overall design phase.

What have I learned? What do I need to learn?

- I learned how PixyMon's colour_connected_components mode works and how saturation, lamp settings, and lighting affects the reliability of colour signature detection.
- I need to learn, how to read PixyCam's output data via Arduino Uno's serial communication to use in code and to integrate it with other subsystems later.
- I communicated the findings from the PixyCam setup to the team, reporting to the leader and confirming that the sensor is viable for ball detection under arena lighting.

3.1.3 Entry 3 (21/02)

What am I doing well?

- Me taking early ownership and of the mapping the system architecture with subsystems to components independently. And, producing a mindmap the whole team can use for clear reference going forward.

What is our team doing well?

- Everyone is progressing on their own tasks and assigned work simultaneously without needing unnecessary formal meetings

Why were the task(s) important?

- The task of mapping every component to a subsystem with a defined role will ensure that the whole team has a mutual understanding of the entire system before prototyping begins. Furthermore, it prevents future issues regarding components being overlooked or used for the wrong purpose. Especially during wiring and assembling the mechanical bits.

What could I have done better?

- I could have communicated with subgroup leaders better and set up a virtual meeting to discuss the component-subsystem mindmap for varied input into the making of it.

Is the progress in alignment with the schedule?

- Yes it is, as the system architecture is essentially fully mapped by the very end of Week 1, ensuring that all subgroups have a clear foundation heading into Week 2 and the project.

What have I learned? What do I need to learn?

- I have learned basically the full list of components provided and how they interacted in between subsystems.
- I should now learn the exact Arduino pin assembly/assignment to operate sensors. Furthermore, I communicated the LucidSpark mindmap to the team for future reference so I have not only learned but in a way "taught" some team members.

3.2 Week 2

3.2.1 Entry 4 (23/02)

What am I doing well?

- By taking ownership of PixyCam and troubleshooting independently and resolving the macOS compatibility issue without needing team support I think was something I did really well.
- Also, by contributing meaningfully to the sensor testing plan ahead of Week 3 physical testing, I set the team in a really good position.

What is our team doing well?

- The team formalised a clear workflow and deadline structure early, which sets a strong foundation for managing the increasing workload over coming weeks.

Why were the task(s) important?

- **Selecting the 4 wheel configuration and claw clamp mechanism on this date was a pivotal decision** because, it gave all subgroups a fixed design direction to work from and it prevents wasted effort on redundant work.

- Furthermore, establishing the workflow and deadline structure now also means the team is organised before the workload peaks rather than after.

What could I have done better?

- I could've come to this meeting with a more concrete sensor testing data to share, rather than just a plan. Having early results would have given the team more to work with when making design decisions.

Is the progress in alignment with schedule?

- Yes, because major design decisions were finalised in Week 2 which seems like a relatively normal week but in reality is crucial to get things done in this week. Workflow established, and prelab completed all within Week 2. Moreover, the team is on track heading into the first lab session.

What have I learned, and what do I need to learn?

- What I learned: how to diagnose and resolve PixyCam driver issues (minor thing), and moreover I learnt the value of locking in key design decisions early to unblock parallel workstreams. Additionally, in terms of technical knowledge I learned what each pin means on the sensor and how to integrate it with the Arduino.
- What I need to learn: how to integrate sensor outputs with Arduino motor control logic ahead of/during Week 3 testing. Additionally, I communicated the PixyCam compatibility findings to the team, the prelim sensor setup and simulations, and contributed to the discussion on ball collection mechanism selection.

3.2.2 Entry 5 (26/02)

What am I doing well?

- I am doing well in terms of leadership of my particular subgroup and across subgroups with other sub leaders, especially by taking initiative to test the servo physically on the Arduino rather than relying solely on simulation.
- I organised this with Franklin via a Teams call and we produced a critical finding about the 180 degree ROM limit that directly informs the gripper mechanism design and its constraints that we have to keep in mind.

What is our team as a whole doing well?

- Our decision to start the EDP presentation in Week 2 well ahead of the Week 5 deadline shows good time management by us, furthermore, continuing to track the progress on a Gantt chart is proving to be incredibly helpful.
- Moreover, dividing crazy eights by subsystem ensures that every area of the robot gets its own dedicated set of people to work on and ideate concepts.

Why were the task(s) important?

- The physical servo test was crucial in allowing our discovery of the 180°ROM limit. Had this been found during the build phase it could have caused significant redesign delays in terms of code mapping to the servos (with inputs greater than 180°on the IDE.)
- Further, scaffolding the problem statement and confirming concept generation tools early gave the team plenty of time to start and refine the EDP presentation properly.

What could I have done better?

- Franklin and I should have tested all of the metal gear servos rather than just one to confirm the ROM limit is consistent across all three servos in the kit. Furthermore, we could have also easily completed running the motor and planning to run it via the driver.

Is the progress in alignment with schedule?

- Yes, it is coming along smoothly and up to date as the servo testing is completed, problem statement scaffolded early, and concept generation tools confirmed all within Week 2, this is comfortably ahead of schedule.

What have I learned and what do I need to learn?

- A key thing I learned as mentioned across the reflection was that the metal gear servos have a maximum ROM of 180°, hence the gripper actuation must be designed and coded within this constraint. This not only showed our identification of limitations but how to grapple with them too.
- In terms of what I need to learn, I think I should read up on how to map specific angles for the servo to our own gripper's positions such as fully open, partially open, and closed.
- Moreover, Franklin and I communicated the 180 degree ROM finding to the team so the mechanical subgroup can factor this into the physical gripper design whilst the programming subgroup can code within this constraint.

3.2.3 Entry 6 (27/02)

What am I doing well?

- One thing I did quite well was that I produced a thorough and well thought out Crazy Eights sheet, which covered 8 unique sensor layout and integration concepts with clear labelling and a sort of a "logic structure" for each concept. I believe if I continue this attention to detail throughout the course, I will be rewarded.
- I further identified a preferred concept on my own in regards to my Crazy Eights (tiered sensing zones). And, I noted its strengths relative to the other concepts I drew, this shows analytical thinking beyond just simple sketching on my behalf. Additionally, most other team members did this sort of thinking too.

What is our team as a whole doing well?

- All team members managed to complete their Crazy Eights on time (well ahead of the formal deadline within late W2) and communicated their designs as images via Teams, this shows great accountability on behalf of every team member. And, the fact that everyone followed through on the agreed deadline structure meant that we were well on track for the EDP presentation in W5.

Why were the task(s) important?

- The Crazy Eights was necessary because it forced quick thinking and it prevents the team members from latching onto the first idea that seems like the "best" one.
- Moreover, the fact that each member needed to cover a specific subsystem meant that our concept generation phase would be comprehensive and there would be no area of the robot design is overlooked. In addition, it is the most efficient way to complete this task too.
- Also, the mindmap and morph chart together gave the team a structured way to collate everyone's own ideas into understandable and reasonable concepts for the EDP presentation ahead of Week 3 and 4.

What could I have done better?

- I think that I could've more formally evaluated each of the 8 concepts against design criteria rather than just identifying a favourite under timed conditions (perhaps an evaluation sheet), which would make the reasoning and my final concept selection more rigorous.
- I believe when I made concept 8 (the minimalist face plate) I dismissed it quickly and kind of just duplicated concept 3 due to the time limit. Perhaps it could have been worth going back and seeing whether enclosing components could modularity benefits during the build phase of ours project.

Is the progress in alignment with schedule?

- Yes, for sure, because everyone's Crazy Eights were completed by the set deadline, the mindmap is being populated and on track to completion. The morph chart will be fully produced by start of week 3. All of these, are keeping the team well on track and even a bit ahead of schedule for concept finalisation and creating our EDP scripts in Week 3.

What have I learned and what do I need to learn?

- I believed that I learned the value of timed concept generation in forcing out quick and even unconventional solutions. This mitigates the outcome of "just the most convenient and *the best* idea being produced." And, I learned that the pan servo camera and tiered sensing zone concepts has the best efficacy for real world detection cases.

- I personally feel that I need to learn exactly *how* the tiered sensing zone logic would translate into actual Arduino code. This is of course for Week 5 and beyond but I would be interested in specifically learning how sensor zones would lead to different signals in the locomotion and retrieval systems.
- In terms of what I communicated, I sent our Teams channel a PDF of my sensor integration Crazy Eights. Thereby contributing to the team and everyone's ideas too. All of which will feed into the final three concepts.

3.3 Week 3

3.3.1 Entry 7 (02/03)

What am I doing well?

- One good thing I am doing is actively contributing to the refining of the problem statement and bringing sensor integration knowledge into the group discussion. Putting us ahead of schedule. Furthermore, **while I am leading** on the technical side of things. I am following along with Jesse and Yusuf's plan for the problem statement and contributing as a team player too.
- The fact that I completed sensor testing independently and producing a code with a working median filter solution for the sensors and bringing this info to the group.

What is our team doing well?

- The team came to the group meeting having all completed their Crazy Eights on time, which meant the session was productive rather than catching up on missed work
- That we sought feedback from both the mentor and lecturer. This proactively shows that our team is serious about hitting the mutually agreed upon HD criteria rather than just submitting something adequate and passing.

Why were the task(s) important?

- We thought that getting the problem statement right early was critical because it anchors the entire EDP presentation based on our reading of the rubric. Because, we figured that it not only is a requirement of the EDP but it actually ensures that we're actually solving the right specific problem.
- Furthermore, sensor testing early on in Week 3 before compliance testing validated the sensor testing planning that I made in Week 2 and it also gave my subgroup and the whole team something to work with/understand going into the integration phase in W5.

What could I have done better?

- I think I really should have tested both IR sensors rather than just the one to confirm that there is consistency, since both will be used in the final design.

- I should've also started drafting concept sketches earlier rather than waiting until the problem statement is fully locked in, because I believe that some parallel progress is always good.

Is the progress in alignment with schedule?

- For sure, I would even say ahead of schedule. Because, I had the sensor testing completed, we all had our Crazy Eights reviewed, the mindmap on track to completion, and the problem statement drafted and presented to Wendy and Brielle all within Week 3.
- Our action item of three concept solutions are targeted by end of Week 3 which is on track to completion

What have I learned, and what do I need to learn?

- I learned that ultrasonic sensors produce messy inaccurate readings in cluttered environments that requires some sort of filtering in code. And, that the IR sensors used inverted logic where LOW signals detection rather than HIGH. **Info gotten from Wendy's technical lectures**
- I strongly think I need to learn how to run both sensors simultaneously on the Arduino. I must consult the electrical subgroup for this.
- I communicated to my team, the sensor testing findings and the codes and the coding process via Teams. And, I contributed to the group problem statement **as a team player** in the editing process by giving my input.

3.3.2 Entry 8 05/03

What am I doing well?

- Taking ownership of the concept sketching and annotation responsibility, which is a direct contribution to the EDP presentation quality
- Contributing meaningfully to the concept synthesis discussion by drawing on sensor integration knowledge from the Crazy Eights session

What is our team as a whole doing well?

- The team effectively synthesised a large amount of ideation output across six individual Crazy Eights, a full mindmap, and a morph chart into three coherent base concepts in a single meeting
- Assigning clear ownership of next steps immediately after the meeting keeps momentum going without ambiguity

Why was/were the task(s) important?

- This meeting was the convergence point of all the divergent ideation done in Week 2 – without it the team would have a lot of ideas but no direction

- Locking in three base concepts this week gives enough time for sketching, Pugh Matrix scoring, and script writing before the Week 5 presentation

What could I have/can I do better?

- Could have come to this meeting with preliminary sketch concepts already drafted, which would have made the discussion more concrete and sped up the finalisation process

Is the progress in alignment with schedule?

- Yes – three base concepts defined, ownership assigned, and Pugh Matrix scheduled for start of Week 4, all on track for the Week 5 presentation deadline

What have I learned, what do I need to learn, and what have I taught/communicated?

- Learned: how to move from divergent ideation tools to a concrete shortlist through structured team discussion, and that the morph chart was the most useful reference point for comparing approaches across subsystems
- Need to learn: how to produce clear and informative annotated engineering sketches that communicate both the design intent and the sensor layout effectively
- Communicated: sensor integration preferences and constraints to the group during concept synthesis, directly influencing which sensor layouts made it into the three final concepts

3.3.3 Entry 9 (07/03)

What am I doing well?

- I strongly believe that me taking initiative as a subgroup leader to coordinate a meeting with the other subgroup leaders over the weekend to lock in the three final concept solutions ahead of schedule was something I did well.
- Also, starting the concept sketches early rather than leaving them to the last minute, was a good choice. Because, to make them good and make the annotations clear and accurate, I would need 2-3 days to refine it to a high quality.

What is our team doing well?

- The fact that all members are independently and keenly doing work and adding onto their EDP sections and scripts without needing constant nagging and coordination from Jesse is a really good thing. This is showing that the subgroup structure we and Brielle established in Week 2 is working really well.

Why were the task(s) important?

- Shortlisting and locking in the three final concept solutions, especially at this point of time is absolutely critical milestone for many reasons. First, it allows me to begin the sketching process, the Pugh Matrix, and the script writing. Because, they all depend on having confirmed concept solutions to work from.

- Furthermore, starting the sketches early gives enough time and headroom for refinement and annotation, which ensures that the quality of the EDP presentation slides is high and presentable.

What could I have done better?

- I should've aimed to fully complete the tasks within the week by coordinating the meeting earlier, rather than carrying the annotation and sketching work into Week 4. Even though the deadline structure technically allows for it to be extended, it would have been better and given me more headroom.
- I could have documented the shortlisting discussion more formally via Teams, so that the reasoning behind each concept selection could be recorded more in detail for this design journal.

Is the progress in alignment with schedule?

- Yes, it is right up to schedule because we have three concept solutions shortlisted, the sketches started, and individual EDP script work is on track to completion. So, I'd say that everything is heading the right direction into Week 4 where the EDP presentation will be solidified and information will be cross checked and shared.

What have I learned and what do I need to learn?

- I have learned that shortlisting from a (really) large pool of ideas is much easier when we have structured concept generation tools like the morph chart as a reference. In my opinion it made trade offs between subsystem approaches much clearer to understand.
- I believe I need to learn how to produce annotated engineering sketches that are clear and well laid out so it can stand alone on a presentation slide without much talking from my side.
- I have communicated the final concept solutions shortlist **and my EDPP script** to the entire team via our unofficial channel on Instagram and also on Teams. This was very beneficial as I could get **feedback** from my team members on what I can improve on and maybe do better next time. **As I firmly believe that as a sub leader and a team player I need to be humble and look for other opinions and input on my work. Rather than being tunnel visioned.**

3.4 Week 4

3.4.1 Entry 10 (09/03)

What am I doing well?

- We completed all three concept sketches with full annotations just in time for the Monday meeting, demonstrating good time management and follow through on tasks carried over from Week 3

- The sketches were clear and detailed enough to anchor the team's Pugh Matrix discussion, which shows the quality of the work was sufficient for its purpose

What is our team as a whole doing well?

- We all completed the Pugh Matrix collaboratively and reached a clear consensus on Concept 3, showing that the structured evaluation process worked well and that all members were engaged in the decision
- Everyone came to the meeting with their EDP sections progressed enough to brief the team, which kept the session productive.

Why were the task(s) important?

- Completing the concept sketches was essential – without annotated visuals the Pugh Matrix discussion would have been abstract and harder to evaluate objectively
- The Pugh Matrix is a critical deliverable for the EDP presentation and formally justifies the selection of Concept 3, which is a key part of the concept generation section worth 40% of the presentation mark

What could I have done better?

- Ideally the sketches should have been fully completed in Week 3 rather than carried into Week 4, even though they were finished on time – tighter self imposed deadlines would reduce last minute pressure

Is the progress in alignment with schedule?

- Yes, because the concept sketches were completed, Pugh Matrix done, and final concept selected all in Week 4 with the presentation still a week away, leaving sufficient time for rehearsal and refinement

What have I learned and what do I need to learn?

- I learned that the Pugh Matrix is most effective when the whole team evaluates together rather than independently, as group discussion surfaces trade offs that individuals might miss
- I need to learn how to deliver the concept solutions section confidently within the 2 minute time limit without reading from notes, as the rubric caps marks at 60% for note reading
- I communicated completed concept sketches and annotations to the full team, which directly enabled the Pugh Matrix session and the final concept selection

3.4.2 Entry 11 (12/03)

What am I doing well?

- I personally, by arriving to the meeting with a fully written and timed script ready for review, did something well and up to team expectations as listed in our team contract. This meant that the session could focus on actually refining the scrip rather than writing new bits or inquiring me about missing sections
- I did not take the lead this session as I wanted to contribute as a team member, to the cohesion of the master script beyond just my own section. I reviewed the final script with Jesse leading the final review.

What is our team doing well?

- The fact that we produced a single cohesive master script with slide change directions and speaker cues shows that we were very well prepared. It showed that we have a high level of preparation ahead of the presentation.
- The team as a whole identified that Yizhuo and Bradman's intro and conclusion needed extra support at the start of the meeting. Thus, we addressed this proactively rather than leaving it to last minute.

Why were the task(s)important?

- Having a master script with transitions makes sure that the presentation has a proper smooth delivery rather than six separate people reading their own sections without any cohesion. This is critical for the **presentation component of the marking rubric worth 20% of the EDP mark**
- The fact that we finalised slides now will leave the weekend solely for rehearsal. This mitigates the risk of last minute changes that could throw off our final performance.

What could I have done better?

- I should've begun rehearsing without notes immediately, because the rubric caps marks at 60% for reading from notes, so I had to memorise key points before Monday. This is non negotiable
- I could have prepared transitional sentences for my section handing over to the conclusion in advance rather than writing during the session which would've saved time
- I should have taken lead in some thing during the meeting, However, for this one I decided to solely be a ream player and contributor.

Is the progress in alignment with schedule?

- Yes, and things are looking good. Since we have the master script completed and slides finalised with 3 days remaining before our W5 Monday presentation. This has left us with more than enough time for independent and group rehearsals.

What have I learned and what do I need to learn?

- I have learned through this whole process that a presentation is really solid and strong when transitions between speakers are scripted rather than improvised. Furthermore, that everyone working towards trimming the script passively gives people info on other sections.
- I need to learn how to deliver the concept solutions section naturally from memory, without sounding robotic and staying within my time limit.
- In terms of communications, I simply presented my concept solutions script and slide structure to the team through OneNote.

References